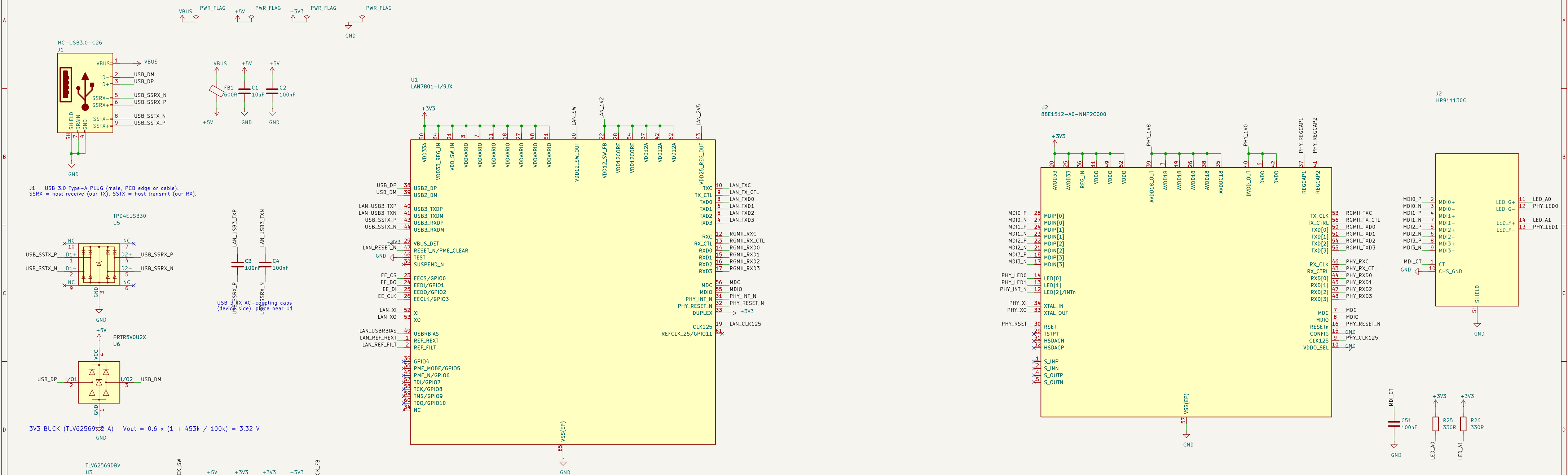


USB 3.0 UPSTREAM + POWER

LAN7801 USB 3 -> RGMII MAC

MARVELL 88E1512 GbE PHY with VCT / TDR

TEST PORT RJ45 1G MAGJACK

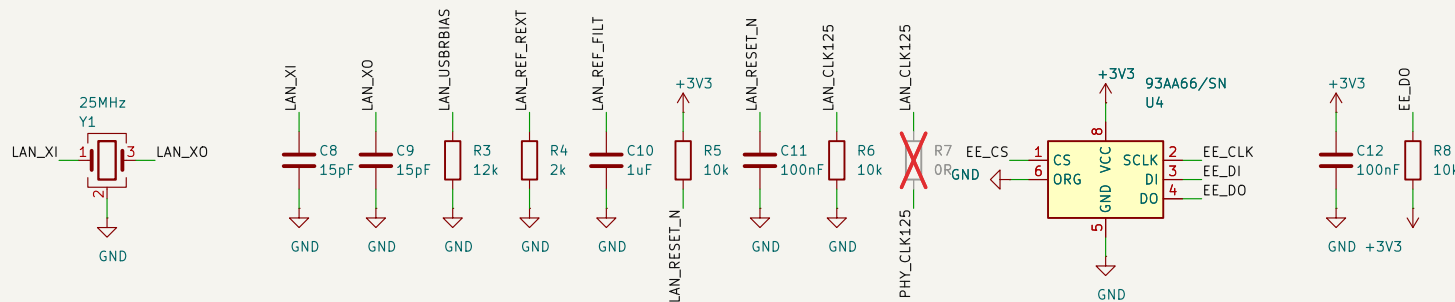


ND-2 TDR PROOF DONGLE – Rev A

First spin: generated from tools/gen_dongle.py, ERC clean, NOT yet reviewed by a human.

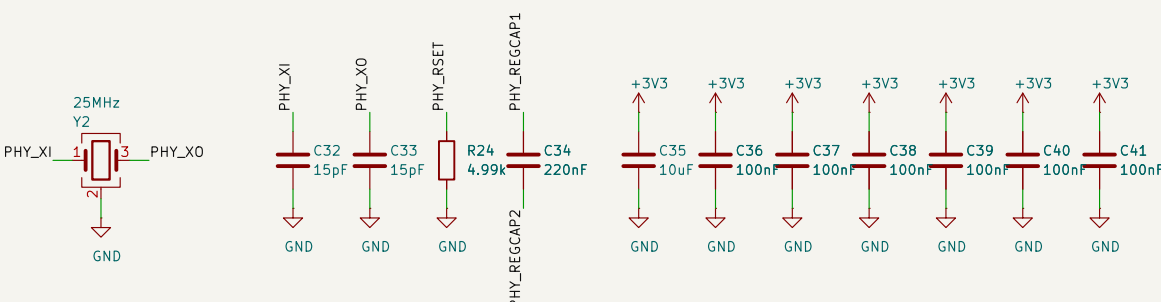
Purpose: prove LAN7801 + BB1512 bring-up and Linux
ethtool --cable-test / --cable-test-tx before building the CM5 carrier.
Bus powered from USB 3 VBUS (VBUS_DET tied to 3V3). Expected load -0.6 A at 3V3.
RGMII delay in exactly one PHY on LAN7801 hybrid mode (MAC adds TXC + RXC delay,
set in F0000). PHY in plain RGMII (phy-mode "rgmii", no PHY delays).
Layout: 90R diff USB / USB 2, 100R diff MDI pairs, 50R single-ended RGMII,
RGMII < 50 m and matched +/-2.5 m, crystals tight to their ICS.
Footprints: LCSC/EasyEDA land patterns in ND2.pretty (QFN pads + EP per datasheet, thermal via added).
Optional: low-cap TVS on the MDI pairs.

LAN7801 support: crystal, bias resistors, reset, CLK125, EEPROM

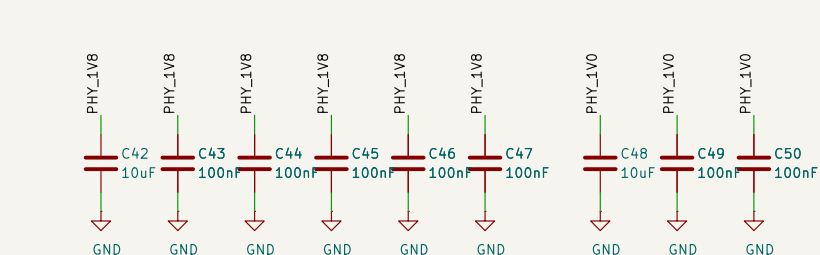


R3 12k, R4 2k: 1 %. U4 93AA66A = 512 x 8 Microwire EEPROM (x8 only): MAC address + HW_CFG (RGMII hybrid mode: MAC TXC + RXC internal delay ON, CLK125_EN = 1, REFCLK25_EN = 0). LAN7801 1V2 switcher, 2V5 LDO cap, 1V2 decoupling

88E1512 support: crystal, RSET, regulator cap, 3V3 decoupling



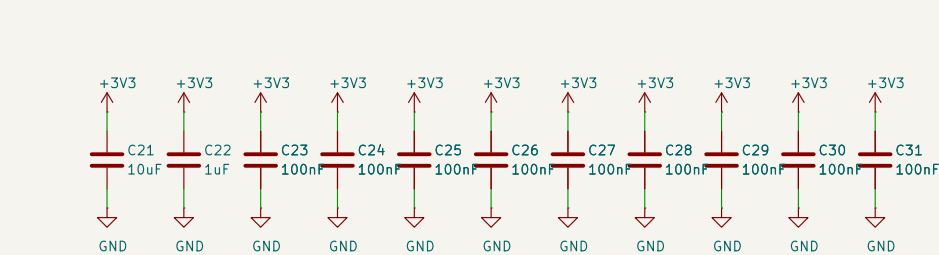
88E1512 regulator outputs: PHY_1V8 (AVDD18 x4 + AVDDC18), PHY_1V0 (DVDD x2),



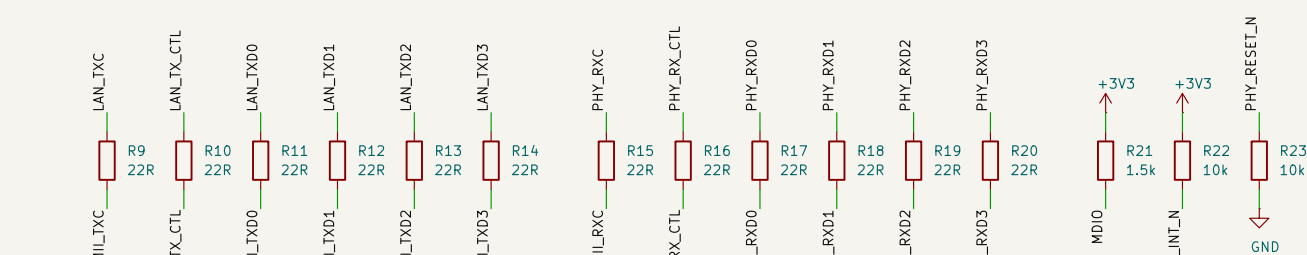
```
R24_4.99k 1 %. XTAL1 is in the 1.8 V AVDDC18 domain (not 3.3 V tolerant): own crystal,
never drive it from LAN7801 REFCLK25.
CONFIG = VSS -> PHYAD = 0, VDDO level 3.3 V, VDDO_SEL = VSS (3.3 V I/O).
Internal switched-cap regulators make PHY_1V8 / PHY_1V0: 220nF REGCAP1-REGCAP2 right at the pins.
PHY_CLK125 reaches LAN7801 CLK125 only if RW (OR, DNP) is fitted:
default: LAN7801 generates CLK125 internally (HW_CFG.CLK125_EN).
```

MDIO...3 -> magjack pairs 1/2, 3/6, 4/5, 7/8 (Auto-MDIX handles swaps).
 HR911130C: common center tap (pin 1) -> 100nF to GND, no bias (voltage-mode PHY). Pin 10 = internal Bob-Smith 1nF/2kV to chassis.
 LEDs are active-low (88E1512 default). LED0 / LED1 functions set in register 3_16.
 LED[2]/INTn -> LAN7801 PHY_INT_N needs LED[2] set to interrupt mode (reg 3_18).
 otherwise lan78xx simply polls the PHY.

LAN7801 3V3 decoupling: 10uF at VDD_SW_IN, 1uF at VDD33_REG_IN, 100nF per VDDVARIO / VDD33A pin



RGMII 22R series terminations (start value): TX group next to U1, RX group next to U2. Length-match all 12 lines



MDIO 1.5k pull-up. PHY_INT_N pull-up. PHY_RESET_N pull-down holds the PHY in reset until the LAN7801 releases it.

First spin - generated, not yet reviewed
LAN7801 USB3-to-RGMII + Marvell 88E1512 PHY (VCT/TDR) + 1G magjack
Network Goblin

Sheet: /
File: tdr-dongle.kicad_sch
Title: ND-2 TDR Proof Dongle

Size: A2	Date: 2026-09-25	Rev: A
KiCad E.D.A. 10.0.5		Id: 1/1